

Chapter 3.1

Control Structures

1. What Are Control Structures?

Control structures are the decision-making and looping mechanisms in Java that control the flow of execution of a program. They let a program:

- Decide what to do — using conditionals (if, if-else, else-if, switch)
- Repeat actions — using loops (for, while, do-while, enhanced for)

2. IF Statement

Executes a block of code only if a condition is true.

```
if (condition) {  
    // statements  
}
```

Example

```
public class IfExample {  
    public static void main(String[] args) {  
        int age = 18;  
        if (age >= 18) {  
            System.out.println("You are eligible to vote.");  
        }  
    }  
}
```

Output: You are eligible to vote.

3. IF-ELSE Statement

Used when there are two possible outcomes — one branch runs if the condition is true, the other if it's false.

```
if (condition) {  
    // executes if condition is true  
} else {  
    // executes if condition is false  
}
```

Example

```
public class IfElseExample {  
    public static void main(String[] args) {  
        int number = 5;  
        if (number % 2 == 0) {  
            System.out.println("Even Number");  
        } else {  
            System.out.println("Odd Number");  
        }  
    }  
}
```

```
    }  
  }  
}
```

Output: Odd Number

4. ELSE-IF Ladder

Used when there are multiple conditions to check, one after another, until one is true.

```
if (condition1) {  
    // block 1  
} else if (condition2) {  
    // block 2  
} else if (condition3) {  
    // block 3  
} else {  
    // default block  
}
```

Example

```
public class ElseIfExample {  
    public static void main(String[] args) {  
        int marks = 75;  
        if (marks >= 90) {  
            System.out.println("Grade A");  
        } else if (marks >= 75) {  
            System.out.println("Grade B");  
        } else if (marks >= 50) {  
            System.out.println("Grade C");  
        } else {  
            System.out.println("Fail");  
        }  
    }  
}
```

Output: Grade B

5. Nested IF Statement

A nested if is an if statement placed inside another if statement. It's used when checking multiple conditions where the second condition only matters if the first one is true.

```
if (condition1) {  
    // executes if condition1 is true  
    if (condition2) {  
        // executes if both condition1 and condition2 are true  
    }  
}
```

Example

```
public class NestedIfExample {  
    public static void main(String[] args) {  
        int age = 20;  
        String nationality = "Indian";  
    }  
}
```

```

        if (age >= 18) {
            if (nationality.equals("Indian")) {
                System.out.println("You are eligible to vote in India.");
            } else {
                System.out.println("You are not eligible to vote in India.");
            }
        } else {
            System.out.println("You must be at least 18 years old to vote.");
        }
    }
}

```

Output: You are eligible to vote in India.

6. SWITCH-CASE Statement

Executes one block from many choices based on a variable's value. Each case needs a break to prevent fall-through into the next case; default runs when no case matches.

```

switch (variable) {
    case value1:
        // statements
        break;
    case value2:
        // statements
        break;
    default:
        // statements
}

```

Example 1 — Day of the Week

```

public class SwitchExample {
    public static void main(String[] args) {
        int day = 3;
        switch (day) {
            case 1: System.out.println("Monday"); break;
            case 2: System.out.println("Tuesday"); break;
            case 3: System.out.println("Wednesday"); break;
            case 4: System.out.println("Thursday"); break;
            case 5: System.out.println("Friday"); break;
            case 6: System.out.println("Saturday"); break;
            case 7: System.out.println("Sunday"); break;
            default: System.out.println("Invalid day");
        }
    }
}

```

Output: Wednesday

Example 2 — Menu-Based Program (Restaurant Order)

```
import java.util.Scanner;
public class RestaurantMenu {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("----- Welcome to Java Cafe -----");
        System.out.println("1. Coffee - Rs.50");
        System.out.println("2. Tea - Rs.30");
        System.out.println("3. Sandwich - Rs.80");
        System.out.println("4. Burger - Rs.100");
        System.out.println("5. Exit");
        System.out.print("Enter your choice (1-5): ");
        int choice = sc.nextInt();
        switch (choice) {
            // cases 1-4 print the item selected
            case 5:
                System.out.println("Thank you! Visit again!");
                break;
            default:
                System.out.println("Invalid choice! Please select 1-5.");
        }
    }
}
```